



MATERIALS 2
Report
Impact Test and Tensile Test

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Academic Year
Semester V
2025-2026

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Laboratory Practice No 2. Tensile Test

Objectives:

- Perform a tensile test on different carbon steel with aluminum sample.
- Obtain the mechanical properties from the curve stress vs strain for each sample.
- Compare the different sample.

Experiment N°1: Analyze the tensile behavior of carbon steel SS400 with heat treatment sample vs non heat treatment sample

Take a sample of each of the following materials Sample:

- Carbon Steel
- Aluminum

1. Measure the dimensions of each sample. Write the data taken in here.

Samples	dimensions
Carbon Steel	85.4 x 11.9 x 2.7
Aluminum	84.5 x 12.6 x 3.1

2. Build the stress vs strain curve. And obtain the following mechanical properties:

❖ By coding We get the result below

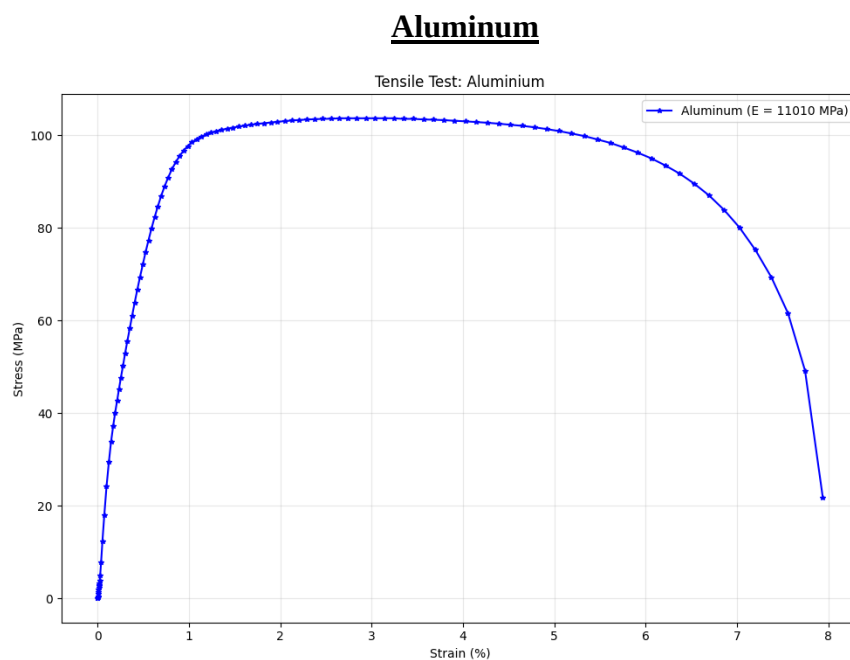


Figure 1. Tensile graph of Aluminum

===== Aluminum Results =====

Ultimate Stress for Aluminum: 103.69 MPa

Fracture Strain for Aluminum: 0.0794

Percent Elongation for Aluminum: 7.94%

E_Aluminum (MPa): 11009.892390201265

Derivative method yield approx: strain aluminum= 0.0073372781065088755 stress aluminum= 89.93999615975451

Carbon Steel

===== Carbon Steel Results =====

Ultimate Stress for Carbon Steel: 467.01 MPa

Fracture Strain for Carbon Steel: 0.2718

Percent Elongation for Carbon Steel: 27.18%

E_Carbon Steel (MPa): 24329.405844382836

Derivative method yield approx: strain Carbon steel = 0.01370023419203747 stress Carbon steel = 339.86111111111121

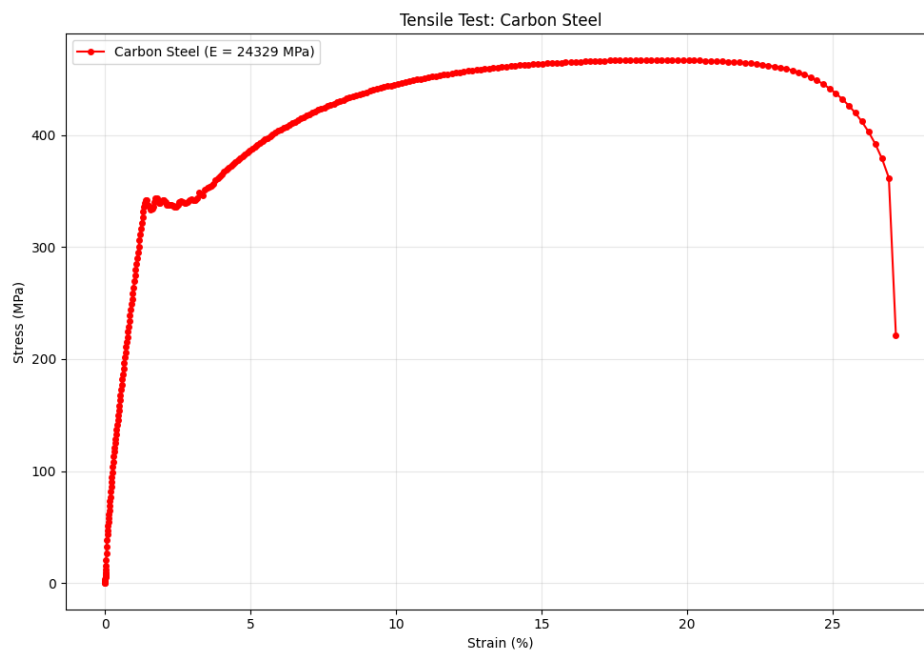


Figure 2. Tensile graph of Carbon steel

- a) Obtain an approximation of the elasticity modulus from the curves.

Samples	E (MPa)
Carbon Steel	24329 MPa
Aluminum	11010 MPa

- b) Calculate the percent elongation

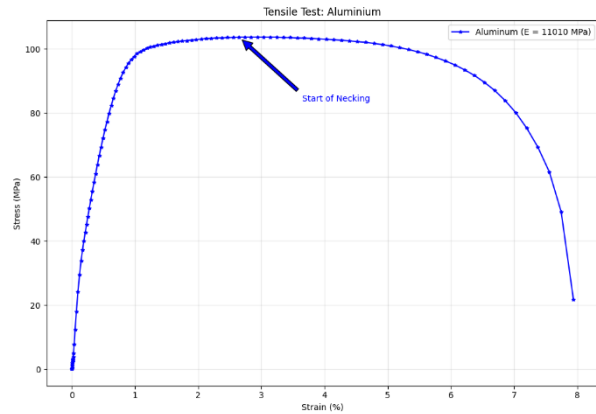
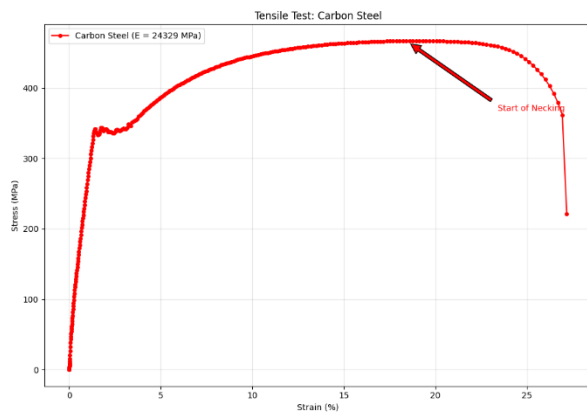
Samples	$\%EL = \left(\frac{l_f - l_o}{l_o} \right) \times 100$
Carbon Steel	27.18 %
Aluminum	7.94 %

c) the fracture strain in each material

Samples	Fracture strain
Carbon Steel	0.2718
Aluminum	0.0794

d) Determine if the materials are ductile or brittle

Since $\%Elongation_{al} = 7.94\%$ and $\%Elongation_{steel} = 27.18\%$, So we can confirm that the materials are **Ductile**. Furthermore, for nicking point of both materials, if it thins out at a specific spot before it breaks, it's ductile. Since both of your materials did this, they are both ductile.



e) Define both the yielding and tensile strengths then determine them for each material.

Samples	Yielding strength (MPa)	Tensile strength (MPa)
Carbon Steel	339.86MPa	467.01Mpa
Aluminum	89.94MPa	103.69MPa

❖ Compare the curve of aluminum and Carbon steel

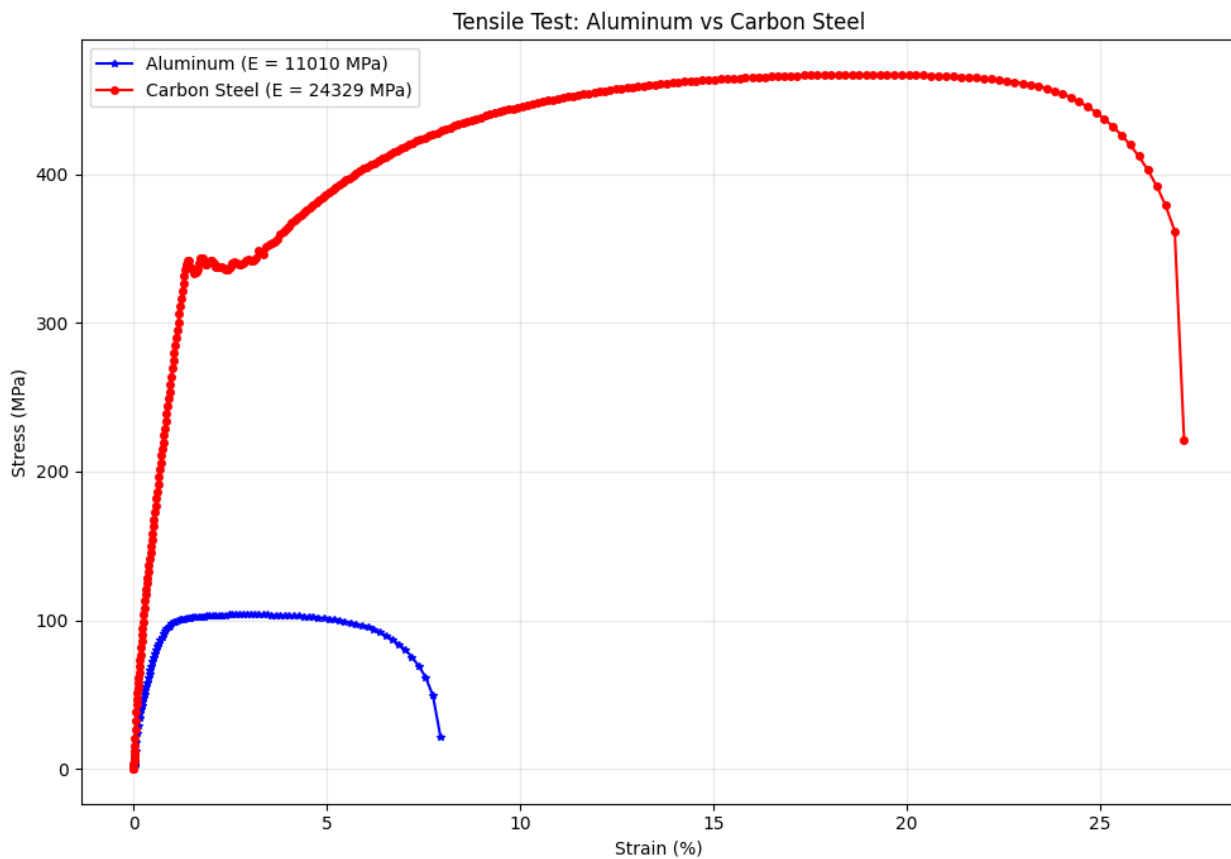


Figure 3. Compare tensile graph of aluminum and Carbon steel

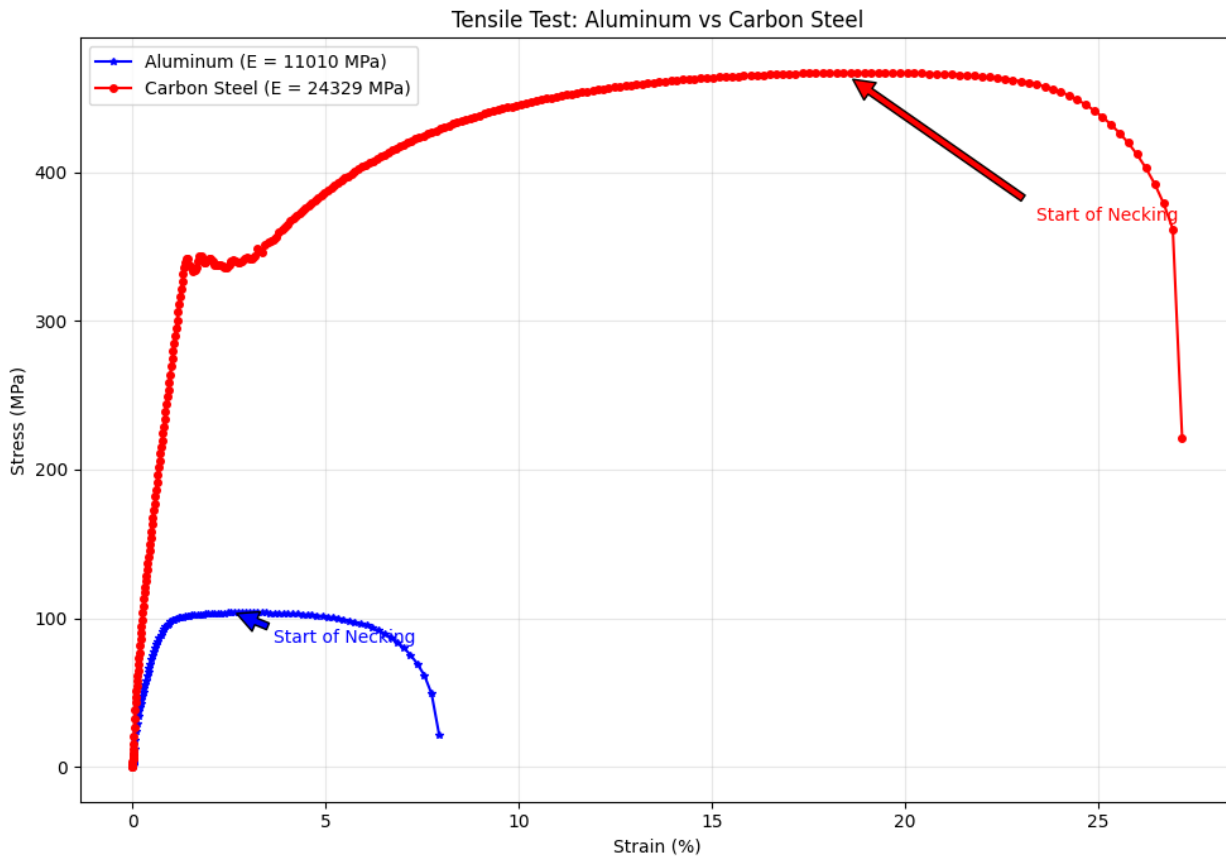
- f) The toughness of the material is the ability to absorb energy up to fracture. Generally, impact tests are performed to obtain the toughness of the material but is possible to obtain an approximation in a tensile test called a static toughness value. Explain:

In a tensile test, a value known as static toughness can be obtained by calculating the total area under the stress-strain curve from the start of loading until fracture. This area represents the energy absorbed per unit volume by the material during deformation. Mathematically, static toughness is given by

$$\text{Static toughness} = \int_0^{\varepsilon_f} \sigma d\varepsilon$$

where σ is the stress and ε_f is the fracture strain

g) Indicate in the curves where necking starts.



h) Conclude the testing result of each sample.

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===== Aluminum Results =====
Ultimate Stress for Aluminum: 103.69 MPa
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===== Carbon Steel Results =====
Ultimate Stress for Carbon Steel: 467.01 MPa
Fracture Strain for Carbon Steel: 0.2718
Percent Elongation for Carbon Steel: 27.18%
E_Carbon Steel (MPa): 24329.405844382836
Derivative method yield approx: strain Carbon steel = 0.01370023419203747 stress Carbon steel = 339.8611111111121
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i) Base on result of yield strength and tensile strength, please find the reference to indicate the type of carbon steel (standard code) and aluminum series of that sample

Based on the tensile test results, the yield strength and ultimate tensile strength of the samples were compared with standard material data.

The carbon steel sample has a yield strength of about 339.86 MPa and an ultimate tensile strength of about 467.01 MPa, which are typical values for low carbon steel. These values closely match those of AISI 1020 carbon steel, so the tested steel is identified as this grade.

The aluminum sample shows a yield strength of about 89.94 MPa and an ultimate tensile strength of about 103.69 MPa. These low strength values are characteristic of commercially pure aluminum. When compared with standard Aluminum Association data, the sample most closely matches the AA 1000 aluminum series (such as AA 1100).

Laboratory Practice N°1: Impact Test

Objective:

- **Perform a Charpy impact test on samples of carbon steel and aluminum.**
- **Obtain the toughness for each material above.**
- **Compare energy absorption of the different materials above.**

Part I. Measure of the Toughness

Toughness is a measure of the ability of a material to absorb energy up to fracture. Specimen geometry as well as the manner of load application are important in toughness determination. For dynamic (high strain rate) loading conditions and when a notch (or point of stress concentration) is presented, notch toughness is assessed by using an impact test.

Two standardized tests, the Charpy (for metals) and Izod (for polymers), were designed to measure the impact energy. For both Charpy and Izod, the specimen is in the shape of a bar of square cross section, into which a V-notch could be machined (see figure below), in which case is called notch toughness. The machine for testing the samples is shown in the figure. The load is applied as an impact from a weighted pendulum hammer that is released from a certain position at a fixed height h . The specimen is positioned at the base as shown. Depending on the type of test: Charpy or Izod, the positioning of the sample differs. Upon release, a knife edge mounted on the pendulum strikes and fractures the specimen at the notch, which acts as a point of stress concentration for this high velocity impact blow. The pendulum continues its swing, rising to a maximum height h' , which is lower than h . The energy absorption, computed from the difference between h and h' is a measure of the impact energy.

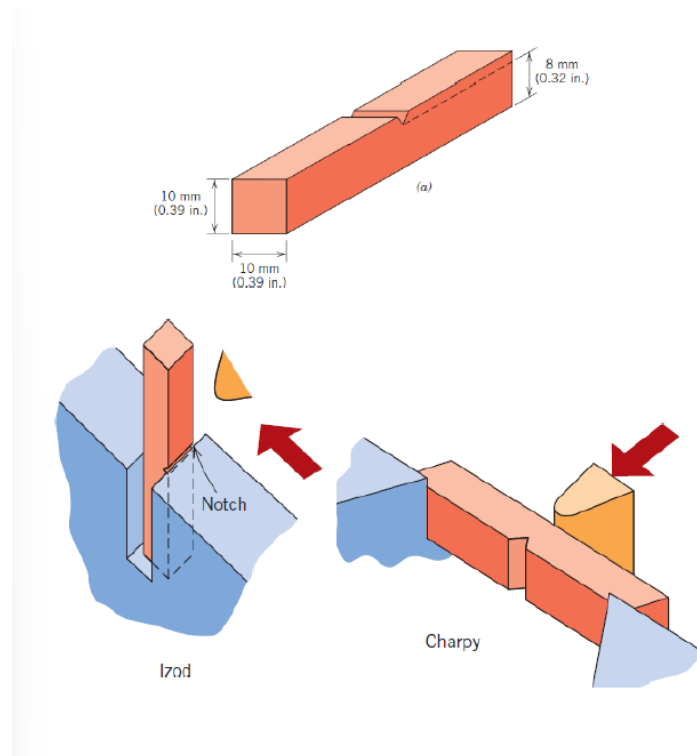


Figure 4. Charpy impact test

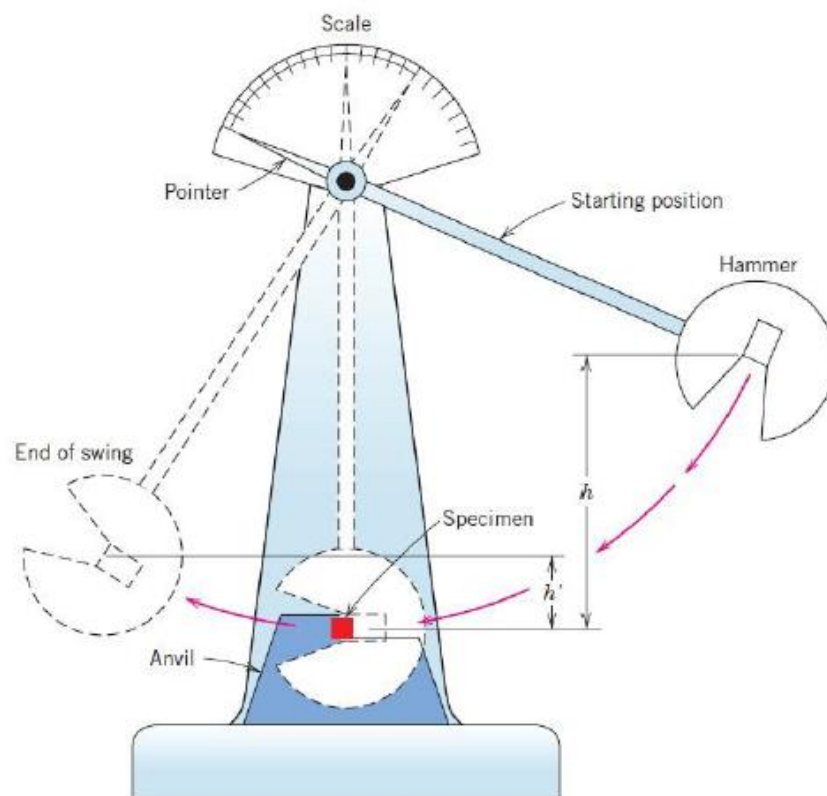


Figure 5. Pendulo Charpy impact test machine

Take a sample of each of the following materials:

- Carbon steel, 2Ps
- Aluminium, 2Ps

1. Measure the dimensions of each sample. Write the data taken in here.

Samples	Dimensions	
Carbon Steel	55.55 x 9.3 x 9.2 mm	56 x 9.2 x 9.75 mm
Aluminium	60 x 10 x 10 mm	59 x 10 x 10 mm

2. Mark the middle point on each sample.

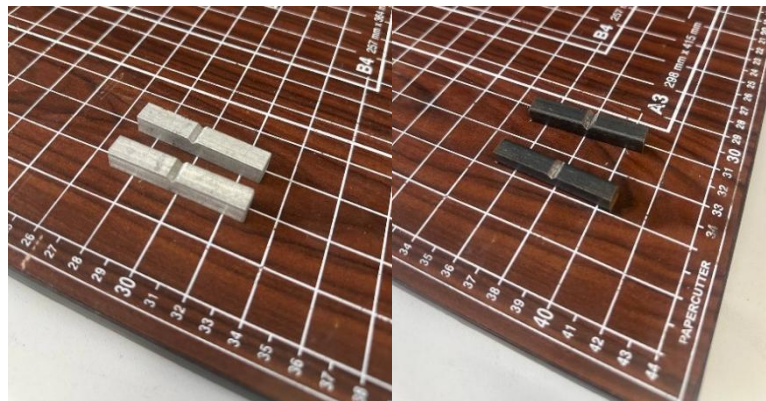


Figure 6. Specimen

3. Place each of the samples in the machine and perform the Charpy Impact test of each material. Do not forget to note the energy measured of each.

- Results of the sample after perform the Charpy Impact test of each material:
+ **Aluminium 1**

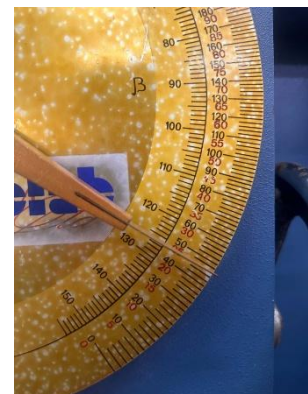
(Before)



(After)



(Close-up)



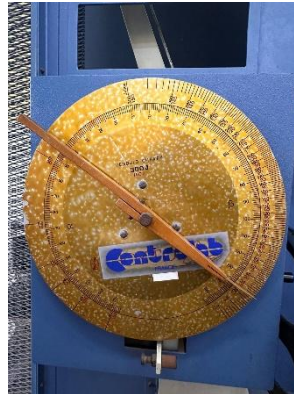
Energy measured: 46 J

+ Aluminium 2

(Before)



(After)



(Close-up)



Energy measured: 24 J

+ Carbon Steel 1

(Before)



(After)



(Close-up)



Energy measured: 185 J

+ Carbon Steel 2

(Before)



(After)



(Close-up)

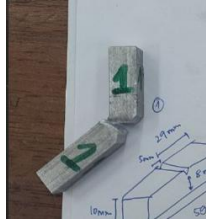


Energy measured: 170 J

4. Place this sample in the machine and do the Charpy Impact Test. Do not forget to note the energy measured.

Notes:

1. Aluminium sample 1: energy measured is 46 J



2. Aluminium sample 2: energy measured is 24 J



3. Carbon Steel sample 1: energy measured is 185 J



4. Carbon Steel sample 2: energy measured is 170J



5. Give the following data for each tested sample:

- Cross-sectional area
- Energy measured
- Resilience (J/cm)

Data of each sample	Value			
	Aluminium 1	Aluminium 2	Carbon Steel 1	Carbon Steel 2
Cross-sectional area	$0.8\text{cm} \times 1\text{cm}$ $= 0.8\text{ cm}^2$	$0.8\text{cm} \times 1\text{cm}$ $= 0.8\text{ cm}^2$	$0.93\text{cm} \times 0.73\text{cm}$ $= 0.6789\text{ cm}^2$	$0.92\text{cm} \times 0.79\text{cm}$ $= 0.7268\text{ cm}^2$
Energy measured	46 J	24 J	185 J	170 J
Resilience (J/cm ²)	57.5 J/cm ²	30 J/cm ²	272.5 J/cm ²	233.9 J/cm ²

6. Classify the materials by increasing ductility

Since we know, ductility is related to a material's ability to absorb energy and undergo plastic deformation before it fractures. Thus, using the resilience value above we can see that samples with higher resilience has more ductility. Ranking the materials by increasing ductility:

Aluminium 2 < Aluminium 1 < Carbon Steel 2 < Carbon Steel 1

7. Explain the difference between Carbon Steel and Aluminium sample.

From the experiment we can conclude the differences such as:

Carbon Steel

- Much higher energy absorption
- Higher resilience values
- Greater plastic deformation before fracture
- Stronger atomic bonding with higher toughness

Aluminium

- Lower energy absorption
- Lower resilience values
- Fractures earlier with less plastic deformation
- Lighter material with lower strength and tough

8. Do you get the same resilience for Carbon Steel and Aluminium with or without an initial notch? Explain.

We won't get the same resilience for Carbon Steel and Aluminium with or without an initial notch because from the formula:

$$\text{Resilience} = \frac{\text{Energy Measured}}{\text{Cross – sectional area}}$$

The Energy Measured will be lowered when there is an initial notch, the impact in the Charpy's sample create stress in a concentrated area around the initial notch, meaning there will be faster crack forming and faster crack propagation. As a result, the specimen absorbs significantly less energy before fracture. Although the notch slightly reduces the cross-sectional area, the reduction in absorbed energy is much greater and dominates the resilience calculation. Meaning, we will get different results with or without an initial notch. We can also conclude that notched specimens exhibit a lower resilience than unnotched specimens for both Carbon Steel and Aluminium.

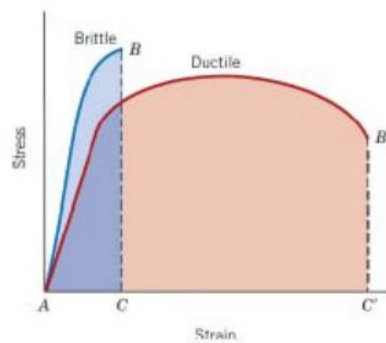
Here are the answers to the theoretical questions based on the stress-strain graph provided (brittle material curve up to point B, ductile material curve extending to B' and C):

Part II. Theoretical questions

Answers:

- 1. Take a look at the stress vs strain data for brittle and ductile materials. Which one shows higher toughness the one with curve ABC or AB'C'?**

It's the ductile material (AB'C'). Toughness is all about how much energy a material can take before it breaks, and you get that from the area under the stress-strain curve. The ductile curve covers a lot more area because it stretches and deforms a lot before snapping. The brittle one (ABC) breaks early, so there's not much area under its curve.



2. So, is ductility tied to toughness?

Absolutely. Ductile materials can be stretched and change shape a lot before they finally break, so they soak up way more energy that's why their toughness is higher. Brittle materials just snap with barely any warning or stretching, so they're not nearly as tough.

3. Now, looking at that same graph, which material is the stiffest?

The brittle one. Stiffness comes down to Young's modulus, which is just the slope of the straight part at the beginning of the curve. The brittle material's curve climbs faster at first, so it has a steeper slope and a higher Young's modulus.

4. Which material is the strongest?

That's the brittle one again. Strength, in this case, means the highest stress a material can handle (that's the ultimate tensile strength—the top of the curve). The brittle material peaks higher at point B than the ductile one does at B', so it wins for strength.

5. Which one will have the biggest percent reduction in area?

That's the ductile material. This number basically tells you how much the material's cross-section shrinks before it breaks. Ductile materials get that necking effect they stretch and thin out a lot at the breaking point, so the percent reduction is big. Brittle materials don't really neck; they just break, so there's barely any reduction in area.